

SpGt Array into 2D subarray

140.1

nums = [7, 2, 5, 10, 8], k = 2.

→ array ko split karo 'K' subarray me
b'w largest subarray sum is minimum.

1st ofc,

$$\text{nums} = [2, 2, 5, 10, 8]$$

Largest element = 10

So, $\sqrt{\text{standard}} = 10.$

ab Largest possible answer:

↳ Sum of entire nums

$$7 + 2 + 5 + 10 + 8 = 32,$$

$$-A \cdot \overbrace{\text{end} = 32}^{\text{end} = 32}$$

i.e 10, 11, 12, ... , 32
 ↑ ↑
 start end.

$$\therefore \text{mid} = s + (e - s) / 2$$

mid = 21

Now, Can I split the array into at most 2 subarrays such that each subarrays sum ≤ 21 ? Yes or No.

Here: Greedy check:-
Start,

$$\text{Current sum} = 0$$

$$\text{sub arrays} = 1$$

$$\text{nums} = [7, 2, 5, 10, 8]$$

Now, add 7.

$$\text{Current sum} = 7.$$

, add 2.

$$\text{Current sum} = 9$$

add 5

$$\text{current} = 14$$

add 10

$$\text{current sum} = 24. \quad \times \text{ not possible.}$$

because $24 > 21 \quad \times$

$$\text{So, subarray} = [7, 2, 5]$$

New subarrays:-

$$[10]$$

add 8

↓

$$10 + 8 = 18 \leq 21 \text{ possible} \rightarrow$$

∴ total 2 subarrays. $\times$ we need.
A10.